

Probability and Computing

Michaelmas Term 2023

Exercise Sheet 6

1. Recall the following Markov chain for shuffling n cards from Problem Sheet 5. In each step, we pick a card uniformly at random, out of the n cards. Then we pick another card independently and uniformly at random, out of the n cards. We exchange the positions of the two cards. Upper bound the mixing time of this Markov chain. In particular, prove that $\tau(\varepsilon) = O(n^2 \log(1/\varepsilon))$.
2. Consider the Hard-Core Gibbs Measure from the lecture notes. λ is a positive real number and $G = (V, E)$ is a graph. For this problem, assume that $0 < \lambda < 1/2$ and that each vertex of G has at most 3 neighbours. $\mathcal{I}(G)$ is the set of independent sets of G . The state space is $\Omega = \mathcal{I}(G)$. The Gibbs probability of a state $I \in \Omega$ is $\pi_I = \lambda^{|I|}/Z(G)$, where $Z(G) = \sum_{I \in \mathcal{I}(G)} \lambda^{|I|}$. We'll consider the Markov chain with state space Ω where the transition from $X_t \in \mathcal{I}(G)$ to X_{t+1} is given as follows.

- With probability $1/2$, $X_{t+1} \leftarrow X_t$
- With probability $1/2$, do the following.
 - Choose v u.a.r. from V
 - $X^+ \leftarrow X_t \cup \{v\}$
 - $X^- \leftarrow X_t \setminus \{v\}$ (* this is in $\mathcal{I}(G)$ since X_t is *)
 - If $X^+ \notin \mathcal{I}(G)$ then $X_{t+1} \leftarrow X_t = X^-$
 - Else
 - * With probability $\lambda/(1 + \lambda)$, $X_{t+1} \leftarrow X^+$
 - * With probability $1/(1 + \lambda)$, $X_{t+1} \leftarrow X^-$.

Show that the unique stationary distribution of the chain is the Gibbs measure (so the stationary probability of each $I \in \Omega$ is π_I), then use path coupling to show that it is rapidly mixing.

3. Let n equidistant points be marked on a circle. Without loss of generality, we think of the points as being labelled clockwise from 0 to $n - 1$. Initially, a wolf begins at 0, and there is one sheep at each of the remaining $n - 1$ points. The wolf takes a random walk on the circle. At each step it moves with probability $1/2$ to its clockwise neighbour and with probability $1/2$ to its anticlockwise neighbour. At each visit to a point the wolf eats a sheep if it is still there. (The sheep don't move.) Which sheep is most likely to be the last eaten?
4. Consider the random walk on the line. For every positive integer i , X_i is chosen uniformly at random from $\{-1, 1\}$. Let $Z_0 = 0$ and $Z_i = \sum_{j=1}^i X_j$. We showed in the lectures that the sequence (Z_j) is a martingale with respect to the sequence (X_i) . Now define $Y_0 = 0$ and, for $i \geq 1$, define $Y_i = \sum_{j=1}^i Z_{j-1}(Z_j - Z_{j-1})$. Show that the sequence (Y_i) is a martingale with respect to the sequence (X_i) .

5. Consider the gambler's ruin problem, where now the independent games are such that the player either wins one dollar with probability p satisfying $0 < p < 1/2$ or loses one dollar with probability $q = 1 - p$. The player continues until the player either loses k dollars or wins n dollars. Let X_i be 1 if the player wins the i 'th game and -1 otherwise. Let $S_i = X_1 + \cdots + X_i$ be the player's net winning after i games. Let T be the stopping time when the player finishes playing. $\mathbb{E}[T]$ is bounded — you may assume this without proof.
- (a) Show that $M_i = (q/p)^{S_i}$ is a martingale with respect to the sequence (X_i) .
 - (b) Determine the probability that the gambler wins n dollars before losing k dollars (as a function of q , p , n and k).
 - (c) Show that $Y_i = S_i + (q - p)i$ is a martingale with respect to the sequence (X_i) .
 - (d) Determine $\mathbb{E}[Y_T]$ and use it to determine $\mathbb{E}[T]$.
6. (Bonus problem: This is a relatively easy question, but you may need to read ahead into Section 5.3 to solve it. It is fine to submit your solutions without it.)
- Let G be an n -vertex regular graph of degree d . We color its vertices red or blue independently and uniformly at random.
- (a) What is the expected number of monochromatic edges (as a function of n and d)?
 - (b) Show that the actual number of monochromatic edges is concentrated around its expectation.